

Measures of Deviation

(Dispersion/Variability)

Introduction

While measures of central tendency tell us about the "center" of data, **measures of deviation** (also called dispersion or variability) tell us how **spread out** the data is. Two datasets can have the same mean but very different spreads!

Example:

- Dataset A: 10, 10, 10, 10, 10 (Mean = 10, No spread)
- Dataset B: 0, 5, 10, 15, 20 (Mean = 10, Large spread)

The main measures of deviation are: **range**, **variance**, **standard deviation**, and **interquartile range (IQR)**.

1. Range

Definition

The **range** is the difference between the maximum and minimum values.

Formula

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

Example

Find the range of: 12, 18, 15, 22, 9, 25, 14

Maximum = 25
Minimum = 9
Range = 25 - 9 = 16

When to Use

- ✓ For a **quick, simple** measure of spread
- ✓ When you need the **total span** of data
- ✓ For initial data exploration

Limitations

- ✗ **Highly sensitive to outliers** (one extreme value changes everything)
- ✗ Ignores all middle values
- ✗ Not useful for statistical inference

2. Variance

Definition

Variance measures the average squared deviation from the mean. It quantifies how far data points spread from the average.

Formulas

Population Variance (σ^2):

$$\sigma^2 = \Sigma(x - \mu)^2 / N$$

Sample Variance (s^2):

$$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$$

Where:

- x = each data value
- μ (or $\bar{x}$) = mean
- N (or n) = number of values
- $(n-1)$ = degrees of freedom for sample

Example

Calculate variance for: 4, 8, 6, 10, 7

Step 1: Find the mean

$$\bar{x} = (4 + 8 + 6 + 10 + 7) / 5 = 35 / 5 = 7$$

Step 2: Find deviations from mean

$$4 - 7 = -3$$

$$8 - 7 = 1$$

$$6 - 7 = -1$$

$$10 - 7 = 3$$

$$7 - 7 = 0$$

Step 3: Square the deviations

$$(-3)^2 = 9$$

$$(1)^2 = 1$$

$$(-1)^2 = 1$$

$$(3)^2 = 9$$

$$(0)^2 = 0$$

Step 4: Sum the squared deviations

$$9 + 1 + 1 + 9 + 0 = 20$$

Step 5: Divide by $(n-1)$ for sample variance

$$s^2 = 20 / (5-1) = 20 / 4 = 5$$

Key Properties

- Always non-negative (≥ 0)
- Variance = 0 means no variability (all values identical)
- Units are squared (e.g., if data is in cm, variance is in cm^2)
- Larger variance = more spread

3. Standard Deviation

Definition

Standard deviation (SD) is the square root of variance. It's the most commonly used measure of spread.

Formulas

Population SD (σ):

$$\sigma = \sqrt{[\sum(x - \mu)^2 / N]}$$

Sample SD (s):

$$s = \sqrt{[\sum(x - \bar{x})^2 / (n - 1)]}$$

Example

Using the variance from the previous example:

$$\text{Variance } (s^2) = 5$$

$$\text{Standard Deviation } (s) = \sqrt{5} \approx 2.24$$

Why Use SD Instead of Variance?

Same units as original data!

- Data in cm → SD in cm (not cm²)
- Makes interpretation easier
- More intuitive to understand

Interpreting Standard Deviation

Small SD: Data points cluster close to the mean

Data: 10, 11, 10, 9, 10
Mean = 10, SD $\approx$ 0.71 (very small spread)

Large SD: Data points spread far from the mean

Data: 0, 5, 10, 15, 20
Mean = 10, SD $\approx$ 7.91 (large spread)

Empirical Rule (for Normal Distributions)

- 68% of data within 1 SD of mean
- 95% of data within 2 SD of mean
- 99.7% of data within 3 SD of mean

When to Use

-  With **normally distributed** data
-  When you need a measure in **original units**
-  For statistical testing and inference
-  Most widely understood measure

Limitations

- ✗ Sensitive to outliers
- ✗ Less useful with skewed distributions
- ✗ Requires more calculation than range

4. Interquartile Range (IQR)

Definition

The **IQR** is the range of the middle 50% of data. It measures the spread of the central portion, ignoring extreme values.

Formula

$$\text{IQR} = Q3 - Q1$$

Where:

- Q1 = First quartile (25th percentile)
- Q3 = Third quartile (75th percentile)

How to Calculate

Step 1: Order the data

Step 2: Find the median (Q2)

Step 3: Find Q1 (median of lower half)

Step 4: Find Q3 (median of upper half)

Step 5: Calculate $\text{IQR} = Q3 - Q1$

Example

Find IQR of: 3, 5, 7, 8, 12, 15, 18, 20, 24

Data is already ordered: 3, 5, 7, 8, 12, 15, 18, 20, 24

Median (Q2) = 12

Lower half: 3, 5, 7, 8

$Q1 = (5 + 7) / 2 = 6$

Upper half: 15, 18, 20, 24

$Q3 = (18 + 20) / 2 = 19$

$IQR = Q3 - Q1 = 19 - 6 = 13$

Identifying Outliers with IQR

Outlier boundaries:

Lower boundary = $Q1 - 1.5 \times IQR$

Upper boundary = $Q3 + 1.5 \times IQR$

Any values outside these boundaries are potential outliers.

When to Use

- ☒ With **skewed** data
- ☒ When data has **outliers**
- ☒ For **robust** measure of spread
- ☒ In **box plots**

Advantages

- **Resistant to outliers** (uses middle 50% only)
 - Works well with any distribution
 - Easy to visualize
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5. Comparing Measures of Spread

Quick Reference Table

Measure	Formula	Pros	Cons	Best Use
Range	Max - Min	Simple, quick	Very sensitive to outliers	Initial exploration
Variance	Average of $(x - \bar{x})^2$	Uses all data	Squared units, hard to interpret	Statistical calculations
Standard Deviation	$\sqrt{\text{Variance}}$	Uses all data, original units	Sensitive to outliers	Normal distributions
IQR	Q3 - Q1	Resistant to outliers	Ignores extreme values	Skewed data, outliers

Relationship Between Measures

For the same dataset:

- Range is always $\geq$ IQR
- Variance = (Standard Deviation)²
- Larger values of any measure = more spread

6. Real-World Applications

Scenario 1: Test Scores

Data: 85, 88, 90, 92, 95, 97, 98

Mean = 92.1
SD = 4.53 (small - scores are consistent)

Interpretation: Low SD means students performed similarly. Teacher can proceed with confidence.

Scenario 2: Stock Prices

Stock A: Mean = \$100, SD = \$2 (stable)

Stock B: Mean = \$100, SD = \$15 (volatile)

Interpretation: Both stocks average \$100, but Stock B has much higher risk (variability).

Scenario 3: Manufacturing Quality

Widget lengths: Mean = 10cm, SD = 0.05cm

Quality Control Rule: Parts must be within 2 SD of mean

- Acceptable range: $10 \pm 2(0.05) = 9.9\text{cm to } 10.1\text{cm}$
- Parts outside this range are defective

Scenario 4: Income Distribution

Data: \$20K, \$25K, \$30K, \$35K, \$40K, \$500K (outlier!)

Mean = \$108K (misleading due to outlier)
SD = \$190K (inflated by outlier)
IQR = \$15K (better representation)

Interpretation: IQR is more useful here because the billionaire outlier doesn't affect it.

7. Sample vs. Population

Population (Entire Group)

- Use **N** in denominator
- Symbol: σ (sigma) for SD, σ^2 for variance
- Example: All students in the entire school

Sample (Subset of Population)

- Use **n-1** in denominator (Bessel's correction)
- Symbol: **s** for SD, **s²** for variance
- Example: 30 randomly selected students

Why n-1?

- Corrects for bias in sample estimates
- Provides better estimate of population variance
- Standard practice in statistics

8. Common Mistakes to Avoid

✗ Mistake 1: Using range with outliers

 **Problem:** One extreme value makes range meaningless

Data: 10, 12, 14, 15, 100

Range = 90 (misleading!)

IQR = 3 (better representation)

✗ Mistake 2: Confusing variance and standard deviation

Remember:

- Variance is **squared units**
- Standard deviation is **original units**
- $SD = \sqrt{\text{variance}}$

✗ Mistake 3: Thinking low SD always means "good"

Context matters!

- Test scores: Low SD = consistency ✓
- Innovation metrics: Low SD = lack of diversity ✗

✗ Mistake 4: Using wrong formula (n vs n-1)

Sample data: Use $n-1$

Population data: Use N

9. Coefficient of Variation (CV)

Definition

CV compares standard deviation to the mean, expressed as a percentage. Useful for comparing variability across different scales.

Formula

$$CV = (SD / \text{Mean}) \times 100\%$$

Example

Dataset A: Mean = 10, SD = 2

$$CV = (2/10) \times 100\% = 20\%$$

Dataset B: Mean = 1000, SD = 20

$$CV = (20/1000) \times 100\% = 2\%$$

Interpretation: Even though Dataset B has larger SD, it has LESS relative variability (2% vs 20%).

10. Practice Problems

Problem 1

Find the range, variance, and standard deviation of: 5, 8, 10, 12, 15

► [Click for solution](#)

Problem 2

Two machines produce widgets:

- Machine A: Mean = 100g, SD = 5g
- Machine B: Mean = 100g, SD = 15g

Which machine is more consistent?

► [Click for solution](#)

Key Takeaways

1. **Range** = Quick measure, sensitive to outliers
2. **Variance** = Average squared deviation, hard to interpret
3. **Standard Deviation** = Most common measure, same units as data

4. **IQR** = Resistant to outliers, uses middle 50%
 5. Choose measure based on **distribution** and **presence of outliers**
 6. Larger values = **more spread/variability**
 7. Use **n-1** for sample variance/SD, **N** for population
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Additional Resources

- Use the **Statistics Calculator** to automatically calculate all measures
 - Take the **Practice Quiz** to test your understanding
 - Learn about the **Empirical Rule** for normal distributions
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Ready to test your knowledge? Take the interactive quiz to master measures of deviation!

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